Candidate Seat No:________________

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
Bachelor of Computer Applications (B.C.A.)
Semester-II University Examination May-2018
CA 117 Operating System Concepts

Date: 09.05.2018, Wednesday Time: 10.00 a.m. to 01.00 p.m. Maximum Marks: 70

Instructions:

1. **Attempt both sections in separate answer books.**
2. **The figure to the right indicates marks.**
3. **Make a suitable assumption when necessary.**

SECTION-I

Q.1[A] **Column A list the Scheduling Algorithms and B with the characteristics of them. Match "A" with "B".** **[05]**

A	B
(1) Round Robin	(a) Dynamic Priority Mechanism
(2) HRRN	(b) Max(w)
(3) SPN	(c) Constant
(4) FCFS	(d) Good Balance to all processes
(5) Feedback	(e) Min(s)

[B] **Fill in the blanks with appropriate answer.** **[06]**

1. An __________ is a signal to the processor emitted by hardware or software indicating an event that needs attention.
2. A good OS collects the _______ data for various resources and monitor the performance parameters such a response time.
3. There was no Operating System in _________processing era.
4. OS creates a process at the explicit request of another process; the action is referred to as ____________.
5. ___________program switches the processor from one process to another.
6. _______number of transitions are possible in seven state process model.

Q.2 [A] What do you mean by Operating Systems? List out the objectives of it and explain the role of it in **(a)** Program Execution and **(b)** I/O access. **[06]**

[B] Define Process. Explain Process Control Block in details. **[06]**

OR

Q.2 [A] Explain five state process model with all aspects. **[06]**

[B] What is the objective of uniprocessor scheduling? Explain different types of scheduling with process state transitions. Explain the scheduling criteria that are important for scheduling algorithms. **[06]**

Q.3 Answer following questions in detail (Any three) **[12]**

1.Give the answers of the questions based on Shortest Process Next(SPN) scheduling policy in the following example.

Processes	Arrival Time	Service Time
A	0	6
B	3	2
C	4	4
D	6	1
E	7	6

(a) What is the waiting time of Process E?
(b) What is the normalized turn round time of Process C?
(c) What is the finish time of Process D?
(d) What is the turnaround time of Process B?

2. Explain the timesharing Operating Systems in details.
3. Draw the state transition diagram of Seven State process model and explain all states of it.
4. List out the preemptive scheduling algorithms and explain any one with suitable example.
5. Explain any eight process termination reasons.

SECTION-II

Q.4 Attempt the following. **[11]**

1. ___________ is a pattern matching special character used to match number of file name.

a) # c) $
b)* d) ?

2. In UNIX file system, there is a top which serves as the reference point for all files called as ___________.

a) root c) dev
b) bin d) home

3. In UNIX, each device is implemented as a file. True / False

4. The _________ interacts with the machine's hardware and the ___________ interacts with the User.

a) Kernel, Shell c) User, Kernel
b) User, Shell d) Shell, Kernel

5. When you login, UNIX automatically places you in a directory called ___________

a) usr c) bin
b) HOME d) path

6. The file descriptor for standard error is ___________.

a) 2 c) 0
b) 3 d) 1

7. grep is an acronym for ___________.

a) Globally search Relative Export and Print
b) Globally search Regular Expression and Print
c) Global Regular Expression Pattern
d) Globally search Relative Expression and Print

8. To run the process in background __________ symbol is used in UNIX.

a) $	c) !
b) *	d) &

9. Pico editor is written in __________ language.

a) C	c) Java
b) C++	d) PHP

10. __________ is the default mode when user starts vi editor.

a) Insert mode	c) The ex mode
b) Command mode	d) Input text mode

11. The root directory is denoted as _________.

a) /	c) .
b) \	d) $

Q.5 [A] Explain in detail all the features of UNIX operating system. **[06]**

[B] Explain following commands with options and suitable examples: **[06]**

a) ls b) grep c) sort

OR

Q.5 [A] Explain UNIX directory structure with diagram. **[06]**

[B] What is called loop? Explain all the types of loop available in shell script with suitable examples. **[06]**

Q.6 Answer following questions in detail (**Any three**) **[12]**

1. Explain with diagram UNIX architecture.
2. What is vi editor? Explain in details modes of vi editor.
3. Explain in detail operators in redirection with example.
4. Explain emacs and pico editors.
5. Write a shell script to check whether a number is odd or even.